Question			Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
4	*(a)*		[A surface that] absorbs all <u>em</u> radiation falling on it / perfect <u>absorber</u> [and emitter] of em radiation / no body is a better emitter of radiation at any wavelength than a black body at the same temperature	1			1		
	(b)	(i)	Polaris [surface] temperature > Chi Pegasi [surface] temperature (1) Intensity of radiation from Polaris > Chi Pegasi [for all λs] accept Polaris brighter than or more luminous than Chi Pegasi (1) Polaris appears 'blue-white' **<u>and</u>** Chi Pegasi appears 'red-orange' (1) Don't accept they are different colours	3			3		
		(ii)	Peak λ identified (400 nm) (1) Application of Wien's law: $\frac{2.9 \times 10^{-3}}{400 \times 10^{-9}}$ = [7 250 K](1) **Alternative:** $\frac{2.9 \times 10^{-3}}{7\,250}$ seen (1) Reference to peak wavelength on graph (400 nm) (1) **Alternative:** Peak wavelength (400 nm) and 7 250 K used correctly (1) to confirm Wien's constant (1)	1	1		2		
		(iii)	$L = 4.05 \times 10^{-9} \times 4\pi \times (431 \times 9.46 \times 10^{15})^2$ (1) - substitution $L = 8.46 \times 10^{29}$ [W] seen (1)	1	1		2	2	
		(iv)	$A = \frac{8.46 \times 10^{29}}{5.67 \times 10^{-8} \times (7\,250)^4}$ (1) - substitution $A = 5.40 \times 10^{21}$ m^2 (1) $4\pi r^2 = 5.40 \times 10^{21}$ $r = 2.07 \times 10^{10}$ [m] (1) (**ecf** on A)		3		3	3	
			Question 4 total	**6**	**5**	**0**	**11**	**5**	**0**